

Department of Mathematics
IIT Hyderabad

March 13, 2024

Course: MA2130: Complex Variables

Final Test

Time: 120 mins

Max marks: 27

Answer any nine of the following. Each question carry 3 marks. $(9 \times 3 = 27)$

1. Find the square roots of $1 - \sqrt{3}i$ and express them in the rectangular form.
2. Let $f(z) = 2xy + i(x^2 - y^2)$ for all $x, y \in \mathbb{R}$. Find all points $z \in \mathbb{C}$ at which f is analytic.
3. Let $u(x, y) = 2x - x^3 + 3xy^2$ for all $x, y \in \mathbb{R}$. Find all harmonic conjugates of u .
4. Is it true that $\log(i^{\frac{1}{2}}) = \frac{1}{2} \log(i)$. Justify your answer.
5. Find $\int_C f(z) dz$, where $f(z)$ is the branch $z^{-1+i} = \exp((-1+i) \log(z))$, where $z = |z| e^{i\theta}$ with $|z| > 0$, $0 < \arg(z) < 2\pi$.
6. Find all analytic functions $f: \mathbb{C} \rightarrow \mathbb{C}$ satisfying $f(z) = f(z+1) = f(\frac{z}{2} + i)$ for all $z \in \mathbb{C}$.
7. Evaluate the integral $\int_0^\infty \frac{x^2}{(x^2+1)(x^2+4)} dx$ using residues.
8. Represent the function $f(z) = \frac{z+1}{z-1}$ by its McLaurin series and state where the representation is valid.
9. Find the Laurent series representation of the function $f(z) = z^2 \sin(\frac{1}{z^2})$ in the domain $0 < |z| < \infty$.
10. Determine the singularities and the associated residues of the function $f(z) = ze^{\frac{1}{z}}$.

End
